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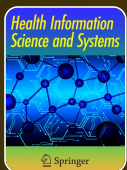
Design and validation of a responsible artificial intelligence–based system for the referral of diabetic retinopathy patients

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Abstract

Diabetic Retinopathy (DR) is a leading cause of vision loss among working-age individuals. Early detection can reduce the risk of vision loss by up to 95%, yet a shortage of retinologists and logistical challenges often delay the DR detection. Artificial Intelligence (AI) systems using Retinal Fundus Photographs (RFPs) present a promising solution. However, their clinical adoption is often hindered by issues such as low-quality data, model biases, learning of spurious features or lack of external validation. To address these challenges, we developed RAIS-DR, a Responsible AI System for DR screening that incorporates ethical principles across the AI lifecycle. RAIS-DR integrates efficient

convolutional models for preprocessing, quality assessment, and three specialized DR classification models. We evaluated RAIS-DR against the FDA-approved EyeArt system on a local dataset of 1046 patients, unseen by both systems. Results are reported for two clinically relevant referral criteria: Referable DR (RDR) and All-Cause Referable (ACR), the latter including low-quality or ungradable images. Evaluations were conducted both per patient and per image. RAIS-DR demonstrated performance improvements in patient-level referral: for RDR, F1-score, accuracy, and specificity increased by 12, 19, and 20%, respectively; for ACR, the corresponding increases were 5, 6, and 10%. RAIS-DR demonstrated equitable performance across demographic subgroups, with Disparate Impact (DI) values between 0.984 and 1.031 and Equal Opportunity Difference (EOD) values near zero. Model explainability helped identify a clinical limitation: false positives were linked to patients with a history of LASER treatment. These findings position RAIS-DR as a robust, reproducible, responsible, and clinically viable solution for DR screening.

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Data Availability

Upon request, data/labels could be available to IBD Lab at <https://fairlac.iadb.org/piloto/retinopatia-diabetica>.

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